25900443

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCHSC200 –
HUMAN NUTRITION AND BIOCHEMISTRY

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **10** questions

Each question carries **2** marks

1. Who is referred as the Indian Reference Man? [K] /[CO1]
   - A**ge:** 19 to 39 years, **Weight:** 65 kg, **Height:** 163 cm, **BMI:** Approximately 20.75, **Physical fitness:** Free from disease and physically fit for moderate activity, - (1 mark).
   - **Occupation:** 8 hours of occupation involving moderate activity, **Rest:** 8 hours in bed, **Light activity:** 4 to 6 hours of sitting and moving about, **Walking/Recreation:** 2 hours of walking, and active recreation or household duties (1 mark)

2. Name one source of protein from animal foods. [U] /[CO1]
   - Egg or Meat/Fish/Milk (2 marks)

3. Define lean body mass. [K] /[CO1]
   - Lean body mass refers to the weight of the body excluding fat. (1 mark)
   - It includes muscles, bones, organs, and body fluids. (1 mark)

4. Give one example of a socio-economic factor affecting food intake. [U] /[CO1]
   - Income level (1 mark)
   - Education or occupation (1 mark)

5. Name two examples for rich sources of carbohydrates. [K] /[CO3]
   - Rice (1 mark)
   - Wheat or Potato (1 mark)

6. What is Protein Efficiency Ratio (PER)? [K] /[CO3]
   - PER is the ratio of weight gain to the amount of protein consumed. (1 mark)
   - It measures the quality of protein in food. (1 mark)

7. Name the ketone bodies. [U] /[CO3]
   - Acetoacetic acid (1 mark)
   - β-hydroxybutyric acid and Acetone (1 mark)
     (if any one – 1 mark, if any two 1.5 marks)

8. What are essential fatty acids (EFA)? [K] /[CO3]
   - EFAs are fatty acids that cannot be synthesized by the body. (1 mark)
   - They must be obtained from the diet, e.g., linoleic acid. (1 mark)

9. Mention two food sources of Folic acid. [K] /[CO4]
   - Green leafy vegetables or citrus fruits (1 mark)
   - Liver or legumes (1 mark)

10. List two deficiency symptoms of Vitamin C. [K] /[CO4]
    - Bleeding gums/joint pain (1 mark)
    - Delayed wound healing or corkscrew-shaped" body hair (1 mark)

11. Write two causes of dehydration. [K] / [CO4]
    - Excessive sweating (1 mark)
    - Diarrhea or vomiting (1 mark)

12. What is the RDA of calcium for pregnant women? [U] / [CO4]

    The RDA of calcium for pregnant women is **1000 mg/day**. (2 marks)

[2x10 = 20]

Part B

Short Essay Type Questions

Answer any **5** questions

Each question carries **4** marks

13. Discuss the importance of formulating Dietary Guidelines for Indians. [U] / [CO1]
    - **Need:** Dietary guidelines help meet nutritional needs, prevent malnutrition, and lifestyle diseases. *(2 marks)*
    - **Importance:** They consider local food habits, climate, socio-economic factors, and cultural diversity to promote balanced and sustainable diets for Indians. *(2 marks)*

14. Write short notes on energy requirement for elderly people. [K] / [CO2]
    - The RDA for energy for elderly people in India is **1700 kcal for sedentary men and 1500 kcal for sedentary women** *(2 marks)*

    Factors influencing: Reduced basal metabolic rate, decreased physical activity, and loss of lean body mass lower energy needs. *(1 marks)*

    - **Recommendation:** Diet should provide adequate energy for basic functions but prevent obesity; focus on nutrient-dense foods. *(1 marks)*

15. Explain why fats yield more energy than carbohydrates. [K] / [CO2]
 Reason: Fats are more reduced compounds containing more hydrogen and less oxygen than carbohydrates. *(2 marks)*

 Result: During oxidation, 1 g of fat yields about **9 kcal**, while 1 g of carbohydrate yields only **4 kcal**. *(2 marks)*

16. List the clinical features of PEM. [K] / [CO3]
 Growth failure
 Muscle wasting
 Edema (especially in kwashiorkor)
 Thin, wrinkled skin (in marasmus)
 Hair changes (discoloration, sparse hair)
 Loss of appetite
 Lethargy and irritability
 Enlarged fatty liver (kwashiorkor) (Each feauture : ½ mark)

17. Explain the significance of deamination. [K] / [CO3]
 Definition: Deamination is the removal of an amino group from amino acids. *(2 marks)*
 Significance: It allows amino acids to be used for energy production or converted to glucose/fat; ammonia formed is converted to urea in the liver. *(2 marks)*

18. Explain the role of Vitamin A in vision. [U] / [CO4]
 Function: Vitamin A forms **rhodopsin**, a pigment in the retina necessary for dim-light vision. *(2 marks)*
 Deficiency effect: Lack of vitamin A leads to **night blindness**, **xerophthalmia**, and other eye disorders. *(2 marks)*

19. Describe the role of kidneys in water balance. [K] / [CO4]
 Function: Kidneys regulate the amount of water excreted in urine by adjusting reabsorption in nephrons. *(2 marks)*
 Mechanism: Under control of **antidiuretic hormone (ADH)**, kidneys conserve water during dehydration and excrete excess during overhydration. *(2 marks)*

[4x5 = 20]

Part C

Essay Type Questions

Answer any **1** questions

Each question carries **10** marks

20. Write an essay on dietary fiber: types, sources and functions. U] / [CO3]

Introduction – 2 Marks

- Dietary fiber refers to indigestible parts of plant foods (cellulose, hemicellulose, pectin, lignin, etc.) that pass through the digestive system largely unchanged.
- It plays a vital role in digestive health and prevention of lifestyle diseases.

Types of Dietary Fiber – 3 Marks

- **Soluble fiber:** Dissolves in water forming gel-like material. Examples: pectin, gums, mucilage.
  - Found in oats, barley, apples, citrus fruits, legumes.
- **Insoluble fiber:** Does not dissolve in water; adds bulk to stool. Examples: cellulose, lignin.

- Found in whole grains, bran, nuts, and vegetables.

Sources – 2 Marks

- Whole cereals and pulses (wheat bran, oats, lentils)
- Fruits and vegetables (apples, carrots, leafy greens)
- Nuts and seeds

Functions – 3 Marks

- Promotes normal bowel movement and prevents constipation.
- Helps control blood sugar and cholesterol levels.
- Aids in weight management by increasing satiety.
- Lowers risk of cardiovascular diseases and colon cancer.

21. Discuss the role of water in maintaining homeostasis in the human body. [U] / [CO4]

Introduction – 2 Marks

- Water forms about 60–70% of body weight.
- It is essential for maintaining internal balance (homeostasis) of temperature, fluids, and electrolytes.

Physiological Roles – 5 Marks

1. **Temperature regulation:** Through sweating and evaporation.
2. **Transport medium:** Carries nutrients, gases, and waste products via blood and lymph.
3. **Solvent function:** Enables biochemical reactions in cells.
4. **Lubrication:** Acts as a lubricant for joints, eyes, and mucous membranes.
5. **Maintains osmotic balance:** Regulates electrolyte and acid-base balance across cells.

Consequences of Imbalance – 2 Marks

- **Dehydration:** Causes fatigue, dizziness, impaired organ function.
- **Overhydration (water intoxication):** Leads to electrolyte dilution, especially sodium.

Conclusion – 1 Mark

- Adequate water intake is vital for maintaining homeostasis and overall physiological stability.

[10x1 = 10]